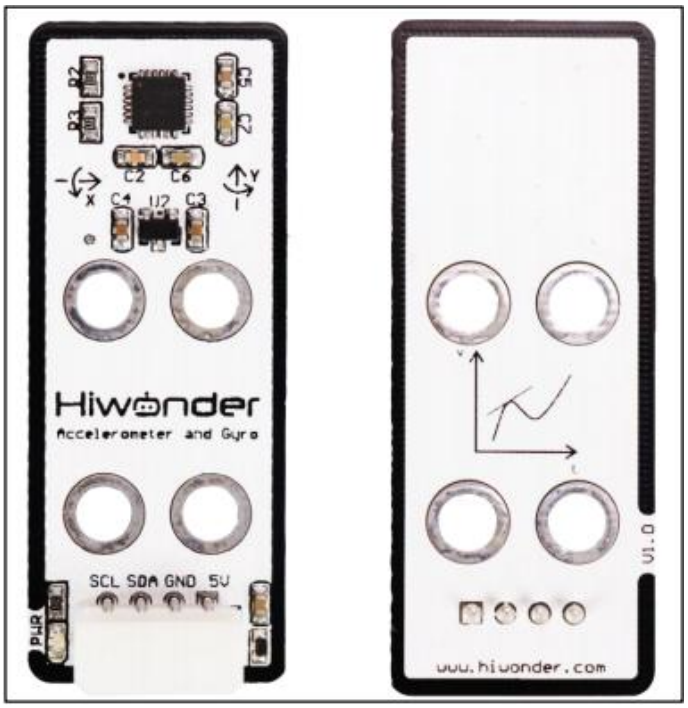
Shenzhen Hiwonder Technology Co,Ltd

Hiwonder Raspberry Pi Development Tutorial

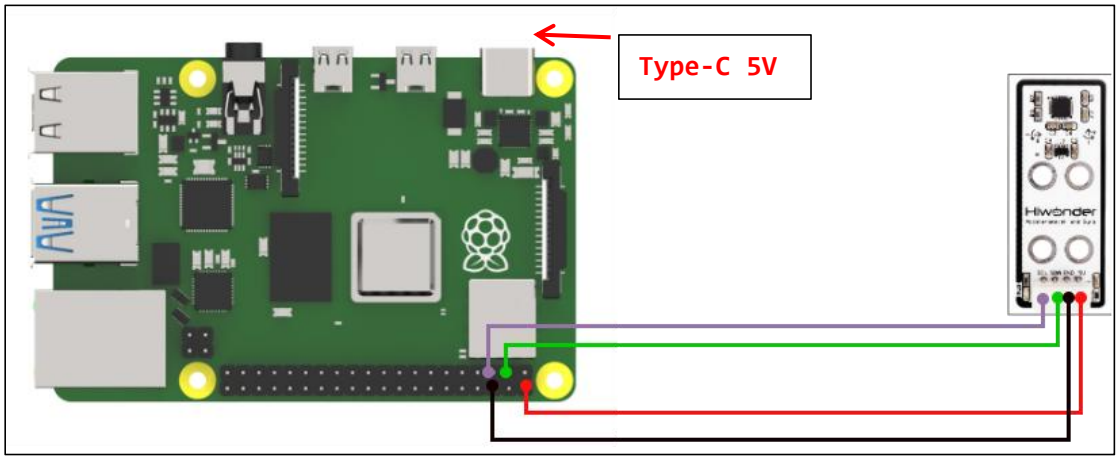
V1.0



1. Getting Ready

1.1 Wiring Instruction

When wiring, you need to connect the 5V, GND, SDA, and SCL pins on the acceleration sensor and their corresponding pins on Raspberry Pi one by one.



Note:

① If you are using the lithium battery provided by our company, please connect its red wire to the positive terminal (+) of the DC port, and its black wire to the negative terminal (-).

② If the lithium battery wires are not connected, do not directly connect them to other wires to avoid a short circuit between the positive and negative poles, which may cause damage to the device.

③ Before powering on, please ensure no metal objects come into contact with the master. This action prevents the pins at the bottom of the master from a short circuit, which may cause damage to the master.

1.2 Environment Configuration

To install NoMachine on your PC, you can access the software package in “2. Software Tools&Program Collections->01 Software Installation Packages-> Remote Desktop Connection Tool”. For detailed instructions on how to use

NoMachine, please refer to the corresponding directory.

Drag the program and the library file “SDK” into the Raspberry Pi system mirror image. Take placing them on the desktop as an example. (Note: Place the library file and the program in the same directory.)

Open the terminal, and input the command “sudo chmod a+x Sensor_Demo/” to switch to the directory where the program is.

```
ubuntu@ubuntu:~/Desktop$ sudo chmod a+x Sensor_Demo/
ubuntu@ubuntu:~/Desktop$
```

2. Case Test

This case uses the Raspberry Pi terminal window to display the detecting status of the acceleration sensor.

2.1 Program Download

(1) Open the terminal, and input the command “cd Desktop/Sensor _Demo/” to switch to the directory where the program is. Press “Enter”.

```
ubuntu@ubuntu:~$ cd Desktop/Sensor_Demo/
ubuntu@ubuntu:~/Desktop/Sensor_Demo$
```

(2) Enter the command “python3 AccelateReadDemo.py” to execute this case program.

```
ubuntu@ubuntu:~/Desktop/Sensor_Demo$ python3 AccelateReadDemo.py
```

2.2 Program Outcome

Place the acceleration sensor in different positions to measure its three-axis angular velocity and acceleration, and view the data change on the serial terminal.

```
ubuntu@ubuntu:~/Desktop/Sensor_Demo$ python3 AccelateReadDemo.py
加速度 (g) - X: 0.00, Y: 3.56, Z: 0.83
陀螺仪 (°/s) - X: 29.96, Y: 499.25, Z: 2.27
加速度 (g) - X: 3.95, Y: 3.58, Z: 0.84
陀螺仪 (°/s) - X: 497.22, Y: 250.14, Z: 479.13
加速度 (g) - X: 3.91, Y: 3.53, Z: 0.87
陀螺仪 (°/s) - X: 495.95, Y: 468.54, Z: 491.45
加速度 (g) - X: 0.32, Y: 3.73, Z: 0.37
陀螺仪 (°/s) - X: 69.09, Y: 249.39, Z: 390.69
加速度 (g) - X: 3.89, Y: 3.40, Z: 1.21
陀螺仪 (°/s) - X: 68.13, Y: 489.92, Z: 448.05
加速度 (g) - X: 3.51, Y: 3.21, Z: 1.12
陀螺仪 (°/s) - X: 478.91, Y: 143.31, Z: 477.07
加速度 (g) - X: 3.11, Y: 3.24, Z: 1.20
陀螺仪 (°/s) - X: 9.88, Y: 122.72, Z: 51.31
加速度 (g) - X: 3.59, Y: 3.54, Z: 0.95
陀螺仪 (°/s) - X: 7.49, Y: 250.14, Z: 36.51
加速度 (g) - X: 3.82, Y: 3.87, Z: 0.71
陀螺仪 (°/s) - X: 97.36, Y: 490.02, Z: 472.63
加速度 (g) - X: 0.02, Y: 0.21, Z: 0.72
陀螺仪 (°/s) - X: 77.88, Y: 127.08, Z: 393.47
```

2.3 Brief Program Analysis

- **Import the Library File**

```
import smbus
import math
import time
```

The required library files “smbus”, “math”, and “time” are imported. The “smbus” is used to communicate with I2C, the “math” provides the function handling math relationship, and the “time” provides the time function.

- **Initialization Operation**

```
#Set I2C bus number, usually as “1”
I2C_BUS = 1

#Set I2C address of MPU6050
MPU6050_ADDR = 0x68

#Register address
MPU6050_ACCEL_XOUT_H = 0x3B
MPU6050_PWR_MGMT_1 = 0x6B

#Initialize I2C bus
bus = smbus.SMBus(I2C_BUS)

#Start MPU6050
bus.write_byte_data(MPU6050_ADDR, MPU6050_PWR_MGMT_1, 0)
```

After the library files are imported, set the I2C bus number and the MPU6050 address to 1 and 0x68 respectively. Set the register address to obtain the X-axis acceleration sensor data, and to control the power mode and

clock setting of the MPU6050 sensor.

Then use the function “smbus.SMBus(I2C_BUS)” to initialize the I2C bus, which is to facilitate the subsequent reading of the sensor data.

Next, use “bus.write_byte_data(MPU6050_ADDR, MPU6050_PWR_MGMT_1, 0)” to state the MPU6050.

- **Read Register Address Data**

```
def read_word(reg):
    high = bus.read_byte_data(MPU6050_ADDR, reg)
    low = bus.read_byte_data(MPU6050_ADDR, reg + 1)
    value = (high << 8) | low
    return value
```

First, use “high = bus.read_byte_data(MPU6050_ADDR, reg)” to read the high-byte data of the register, and then use “high = bus.read_byte_data (MPU6050_ADDR, reg)” to read its low-byte data.

Next, combine the data of these two bytes to return the value.

- **Read Sensor Data**

```
def read_sensor_data():
    accel_x = read_word(MPU6050_ACCEL_XOUT_H)
    accel_y = read_word(MPU6050_ACCEL_XOUT_H + 2)
    accel_z = read_word(MPU6050_ACCEL_XOUT_H + 4)
    gyro_x = read_word(MPU6050_ACCEL_XOUT_H + 8)
    gyro_y = read_word(MPU6050_ACCEL_XOUT_H + 10)
    gyro_z = read_word(MPU6050_ACCEL_XOUT_H + 12)

    return accel_x, accel_y, accel_z, gyro_x, gyro_y, gyro_z
```

Call the “read_word” function and use “MPU6050_ACCEL_XOUT_H” as the register address to return the obtained value to “accel_x” corresponding to the X-axis data of the acceleration sensor.

Similarly, “accel_y”, “accel_z”, “gyro_x”, “gyro_y”, and “gyro_z” represent respectively the acceleration sensor data of the y-axis and z-axis and three axes’ data of the gyroscope sensor.

- **Main Function**

```
def main():
    try:
        while True:
            accel_x, accel_y, accel_z, gyro_x, gyro_y, gyro_z = read_sensor_data()

            #According to the MPU6050 data manual, perform data processing and unit conversion
            accel_x_scaled = accel_x / 16384.0
            accel_y_scaled = accel_y / 16384.0
            accel_z_scaled = accel_z / 16384.0
            gyro_x_scaled = gyro_x / 131.0
            gyro_y_scaled = gyro_y / 131.0
            gyro_z_scaled = gyro_z / 131.0

            #filtering
            print(f"加速度 (g) - X: {accel_x_scaled:.2f}, Y: {accel_y_scaled:.2f}, Z: {accel_z_scaled:.2f}")
            print(f"陀螺仪 (°/s) - X: {gyro_x_scaled:.2f}, Y: {gyro_y_scaled:.2f}, Z: {gyro_z_scaled:.2f}")

            time.sleep(0.1)  #Update data once per second

    except KeyboardInterrupt:
        pass
```

First, call the function “read_sensor_data” to obtain the acceleration and gyroscope data of the three axes. And assign them to 6 variables.

Then, divide the obtained data by their corresponding constants to convert the units, and to reassign the values.

Finally, print these converted data with two decimal places.